OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 23 OF 55

AI-Based Meeting Summarization and Action-Item Extraction Using Natural Language Processing

Domain: Speech & Natural Language Processing | **Module:** project_23_meeting_summarization

OFFICIAL ABSTRACT

Meetings are an important part of academic, business, and organizational activities, where discussions often contain decisions, responsibilities, deadlines, and important follow-up tasks. However, manually recording and reviewing lengthy meeting conversations can be time-consuming, and important action items may be overlooked. The proposed AI-Based Meeting Summarization and Action-Item Extraction System Using Natural Language Processing (NLP) aims to automatically convert meeting conversations into concise and meaningful information that can be easily understood and followed. The proposed system is implemented using Python, Natural Language Processing techniques, and machine learning/deep learning models to process meeting transcripts or speech-to-text content. The system analyzes the conversation, identifies important discussion points, removes unnecessary or repetitive information, and generates a concise summary of the meeting. Instead of simply shortening the entire conversation, the system focuses on extracting information that is useful for understanding the actual outcome of the meeting. The major uniqueness of the proposed system is its action-oriented summarization approach. Along with generating a general meeting summary, the system specifically identifies statements that represent tasks, decisions, responsibilities, and deadlines. For example, when a participant is assigned a particular task during the discussion, the system can recognize the task and associate it with the responsible person and expected completion date. This converts an unstructured conversation into a structured set of actionable information. Another important feature is the ability to distinguish between general discussion and final decisions. Meetings often contain multiple suggestions and opinions before reaching a conclusion. The proposed system attempts to identify the final decisions and important outcomes rather than treating every statement as equally important. This helps users quickly understand what was actually decided and what actions need to be taken after the meeting. The extracted information can be presented as a

structured meeting report containing the meeting summary, key discussion points, decisions, action items, responsible participants, and deadlines. Such a format reduces the effort required to prepare meeting minutes manually and provides participants with a clear reference for future follow-up. The system can also be extended to process meetings from different domains, including project management, education, software development, team discussions, and organizational meetings. The proposed system can further support conversational or real-time meeting environments by processing speech converted into text and continuously identifying important information. With suitable integration, extracted action items can also be connected to task-management or notification systems, helping users remember pending responsibilities and deadlines. Overall, the project demonstrates how NLP and AI can transform lengthy and unstructured meeting conversations into concise, organized, and actionable knowledge. By identifying not only what was discussed but also what was decided, who is responsible, and what needs to be completed, the proposed system provides a practical solution for improving meeting productivity, accountability, and follow-up efficiency.

📌 Project Summary & System Specifications

Project Title	AI-Based Meeting Summarization and Action-Item Extraction Using Natural Language Processing
Specialization / Category	Speech & Natural Language Processing
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5023`
Local Implementation	`project_23_meeting_summarization`